Task-6

Privilege Levels in Linux: Kernel Mode and User Mode

Linux operates with two primary privilege levels - Kernel Mode and User Mode - to ensure system security stability and efficient resource management. These levels define what a process is allowed to do based on its access rights.

User Mode

It is restricted environment where regular applications run. Processes here can't directly access hardware or kernel memory.

Key points:-
- → Running apps like web browsers or editors.
- → Using commands like 'ls or gcc.
- → calling functions like open() or printf() (internally converted to system call)

Kernel Mode

It is the privileged mode where the Linux kernel operates, with full access to all hardware and memory.

Key points:
- Executes critical system functions: process scheduling, memory management, device control
- Handles system calls and hardware interrupts.
- A fault here can crash the system

Ex:-
- Handling read() or write() system calls.
- Managing device drivers and file systems.
- Allocating memory using paging.

Mode Switching:-
- A system call from User Mode triggers a controlled switch to kernel Mode.
- Once the task is complete, control returns to User Mode.